

IT 474 Course Project

2nd Semester 1447

Defensive and Offensive Security Project

Recently, we saw the evolution of internet technologies such as smart cities [1, 2], medical devices [3, 4], and the development of network platforms like cloud computing [5], wireless and mobile [6, 7]. However, these technologies pose security concerns and face many security threats.

There are many types of threats and network attacks, which differ in the degree of complexity and consequences. In teams of **Two** Network Security engineers, select either defensive tool or any type of network attacks and write about the main points: **describe the tool pros and cons/ attack and its severity?** Who was the **developer/ target?** How do you **use the tool/ prevent the attack?** The used tools (**environment setup**). Also, support your presentation with a simulation (**demo**) of the attack, which describes the attack easily.

Important notes:

- The demo should be prepared by yourselves. **NO animation NOR available videos are allowed.**
- **There are two marks for creativity and complex work.**
- **Make sure your presentation includes:**
 - The defensive/ offensive security objectives,
 - An introduction that describes the network platform or environment was selected.
 - Introduction which describes the tool and its pros (and cons) /the attack and its severity,
 - Who was the **provider/ target?** How do you **use the tool/ prevent the attack?**
 - The used tools in the demo (**environment setup**).

References:

- [1] <https://www.helpnetsecurity.com/2019/08/21/cybersecurity-smart-cities/>
- [2] https://www2.deloitte.com/content/dam/Deloitte/de/Documents/risk/Report_making_smart_cities_cyber_secure.pdf
- [3] <https://phoenixnap.com/blog/healthcare-cybersecurity-threats>
- [4] https://www.bsigroup.com/LocalFiles/EN-AU/ISO%2013485%20Medical%20Devices/Whitepapers/White_Paper_Cybersecurity_of_medical_devices.pdf
- [5] <https://phoenixnap.com/blog/best-network-security-tools>
- [6] <https://www.computerweekly.com/tip/Four-mobile-device-security-threats-and-three-tools-to-manage-them>
- [7] <https://auth0.com/blog/ten-mobile-security-threats-and-what-you-can-do-to-fight-back/>

Project Timeline

Phase #	Requirements	End Date
<u>Phase #1:</u> Formulate groups and select a topic.	Use the provided link in the LMS to decide your project topic and write a small description to be approved by the lab team.	Sunday, 29 March Week 11 Through LMS
<u>Phase # 2:</u> Submission and presentation	In this phase, you should: 1. Create a demo that demonstrates the main points in your selected topic. 2. Prepare slides that include the main topics that were explained before. 3. Present your project in class (exact time slots and dates will be announced for each group later). You will have only 10 minutes to present and 5 minutes for the discussion.	Sunday, 3 May Week 16 Through LMS and Lab time